

LAB 2: Build Kernel Image

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1. Q1: Shrink the size of your kernel image

A. Build Kernel

產生 default config.txt 檔

```
make ARCH=arm64 CROSS_COMPILE=aarch64-linux-gnu- bcm2711_defconfig
```

以 config 檔 build kernel

```
make ARCH=arm64 CROSS_COMPILE=aarch64-linux-gnu- Image modules dtbs
```

B. Flash Image

插入 sd card，會看到 device 中有 sdb1、sdb2，分別 mount 到 /fat32 與 /ext4

```
sudo mount /dev/sdb1 mnt/fat32
```

```
sudo mount /dev/sdb2 mnt/ext4
```



在 linux 資料夾下創建掛載點，掛載點下再創建 fat32 文件系統和 ext4 文件系統，分別和 /sdb1 和 /sdb2 掛載。簡單來說，FAT32(sdb1)是用來儲存 kernel image 的，以及配置文件如 config.txt、cmdline.txt 他們是啟動的必須文件。而 EXT4(sdb2)則是 data filesystem，類似於跟目錄儲存系統的文件。

接著在 SD card 上 install kernel 的模組

```
sudo env PATH=$PATH make ARCH=arm64 CROSS_COMPILE=aarch64-linux-gnu-  
INSTALL_MOD_PATH=mnt/ext4 modules_install
```

將 device tree blobs copy 到 SD card 上

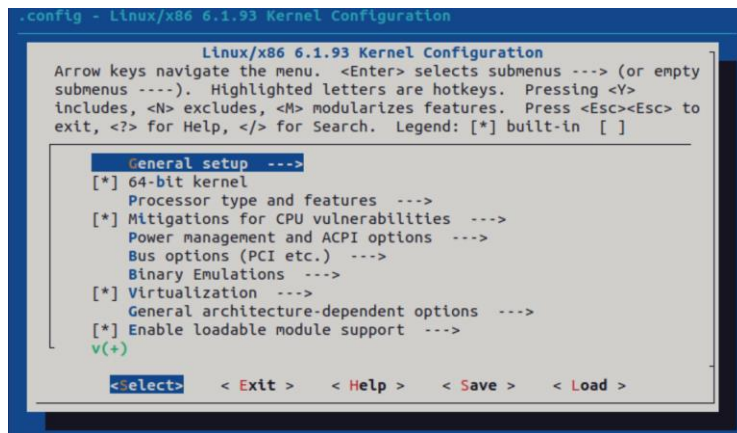
```
sudo cp mnt/fat32/$KERNEL.img mnt/fat32/$KERNEL-backup.img sudo cp  
arch/arm64/boot/Image mnt/fat32/$KERNEL.img sudo cp  
arch/arm64/boot/dts/broadcom/*.dtb mnt/fat32/ sudo cp  
arch/arm64/boot/dts/overlays/*.dtb* mnt/fat32/overlays/ sudo cp  
arch/arm64/boot/dts/overlays/README mnt/fat32/overlays/
```

C. Configure your kernel

透過 menuconfig 修改 kernel option

```
make ARCH=arm64 CROSS_COMPILE=aarch64-linux-gnu- menuconfig
```

會看到如下圖的圖形介面



反覆修改，關閉不需要的功能並重新 make，觀察 kernel size 變化。透過

`ls -lh /boot`

可以看到原先大小為 23MB

```
-rw-r--r-- 1 wang wang 8.9M 七 4 08:05 kernel_2712.img
-rw-r--r-- 1 wang wang 23M 九 21 22:43 kernel8-backup.img
-rw-r--r-- 1 wang wang 23M 九 21 22:43 kernel8.img
```

在這次的實驗中，我關閉的選項有如下

Device drivers

|-----Multimedia Support

|-----Remote Controller Support

|-----LED Support

|-----Sound Card Support

|-----Hardware Monitoring Support

|-----Thermal Driver

|-----Real Time Clock

|-----Auxiliary Display Support

File System

|-----Network File System

Power Management

Security Options

以上關閉的選項的介紹會在下一部分作介紹。可以看到，當關閉以上選項重新 make 並燒錄至 SD card 後，其 kernel size 確實變小。

```
-rwxr-xr-x 1 root root 8.9M 七 4 08:05 kernel_2712.img
-rwxr-xr-x 1 root root 23M 九 24 11:20 kernel8-backup.img
-rwxr-xr-x 1 root root 20M 九 24 11:20 kernel8.img
```

kernel8.img 變為 20MB，而 kernel8-backup.img 存放的則是舊的映像檔做備份。

2. Q2: Benchmark your kernel, revise it, and improve its performance. Rerun the benchmark to prove your performance.

首先，先介紹關閉的選項的功能。以下分別為其介紹:

A. Multimedia Support

此部分涉及支援多媒體的相關設備、圖形處理、影片邊解碼等功能。

B. Remote Controller Support

支援遠端控制器，通常像是鍵盤、滑鼠等。

C. LED Support

管理與控制樹梅派上的 LED 燈，例如 power on、SD card ACT 狀態燈等。

D. Sound Card Support

此部分即為聲卡驅動，提供音頻的輸出。透過 `aplay -l` 列出聲音驅動裝置。Aplay 是一個在 linux 系統中用來撥放音的工具，全名為 Advanced Linux Sound Architecture 簡稱 ALSA。可以看到關閉前後的狀態。

```
pi@raspberrypi:~$ aplay -l
**** List of PLAYBACK Hardware Devices ****
card 0: Headphones [bcm2835 Headphones], device 0: bcm2835 Headphones [bcm2835 Headphones]
  Subdevices: 8/8
    Subdevice #0: subdevice #0
    Subdevice #1: subdevice #1
    Subdevice #2: subdevice #2
    Subdevice #3: subdevice #3
    Subdevice #4: subdevice #4
    Subdevice #5: subdevice #5
    Subdevice #6: subdevice #6
    Subdevice #7: subdevice #7
card 1: vc4hdmi0 [vc4-hdmi-0], device 0: MAI PCM i2s-hifi-0 [MAI PCM i2s-hifi-0]
  Subdevices: 1/1
    Subdevice #0: subdevice #0
card 2: vc4hdmi1 [vc4-hdmi-1], device 0: MAI PCM i2s-hifi-0 [MAI PCM i2s-hifi-0]
  Subdevices: 1/1
    Subdevice #0: subdevice #0
pi@raspberrypi:~$
```

```
pi@raspberrypi:~$ aplay -l
aplay: device_list:274: no soundcards found...
pi@raspberrypi:~$
```

E. Hardware Monitoring Support

用來監控硬體狀態如 CPU 溫度、電壓等功能。

F. Thermal Driver

與溫度有關的驅動，通常用在監控避免過熱。

G. Real Time Clock

簡稱 RTC，為系統提供硬體層面的時間管理。

H. Auxiliary Display Support

輔助顯示支援，和外接螢幕有關，若只會在命令行介面運行的話可以關閉。

I. Network File System

簡稱 NFS，是一個透過網路掛載的共用文件系統，可以由不同的裝置存取。

J. Power Management

電源管理，裝置可以進入不同的電源模式如休眠、睡眠、待機等。

K. Security Options

和安全有關的功能，例如 SELinux、AppArmor。

接著，將進行 benchmark 的部分，我選擇網路上兩個測試工具分別是 Sysbench 和 Geekbench，結果如下：

A. Sysbench

- Before reducing kernel size

-----CPU-----

```
pi@raspberrypi:~$ sysbench cpu run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Prime numbers limit: 10000
Initializing worker threads...

Threads started!

CPU speed:
  events per second: 1774.24

General statistics:
  total time:          10.0003s
  total number of events: 17749

Latency (ms):
  min:                 0.56
  avg:                 0.56
  max:                 0.93
  95th percentile:    0.57
  sum:                 9995.47

Threads fairness:
  events (avg/stddev): 17749.0000/0.00
  execution time (avg/stddev): 9.9955/0.00
```

-----Memory-----

```
pi@raspberrypi:~$ sysbench memory run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Running memory speed test with the following options:
  block size: 1KiB
  total size: 102400MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 24584599 (2457611.26 per second)
24008.40 MiB transferred (2400.01 MiB/sec)

General statistics:
  total time:          10.0001s
  total number of events: 24584599

Latency (ms):
  min:                 0.00
  avg:                 0.00
  max:                 0.05
  95th percentile:    0.00
  sum:                 4848.96

Threads fairness:
  events (avg/stddev): 24584599.0000/0.00
  execution time (avg/stddev): 4.8490/0.00
```

- After reducing kernel size

-----CPU-----

```
pi@raspberrypi:~$ sysbench cpu run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Prime numbers limit: 10000
Initializing worker threads...

Threads started!

CPU speed:
  events per second: 1767.50

General statistics:
  total time:          10.0006s
  total number of events: 17682

Latency (ms):
  min:                 0.56
  avg:                 0.57
  max:                 2.10
  95th percentile:    0.57
  sum:                9994.86

Threads fairness:
  events (avg/stddev): 17682.0000/0.00
  execution time (avg/stddev): 9.9949/0.00
```

-----Memory-----

```
pi@raspberrypi:~$ sysbench memory run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Running memory speed test with the following options:
  block size: 1KiB
  total size: 102400MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 24736634 (2472814.78 per second)
24156.87 MiB transferred (2414.86 MiB/sec)

General statistics:
  total time:          10.0001s
  total number of events: 24736634

Latency (ms):
  min:                 0.00
  avg:                 0.00
  max:                 1.11
  95th percentile:    0.00
  sum:                4874.52

Threads fairness:
  events (avg/stddev): 24736634.0000/0.00
  execution time (avg/stddev): 4.8745/0.00
```

	Before	After
CPU	1774.24	1767.50
statistic(events/sec)		
Memory	2400.01	2414.86
statistic(MB/sec)		

透過 sysbench 分析 CPU 與 memory 的表現。在 CPU 的部分以 events per second 為主要指標，在分析每秒 CPU 可以完成的 event 數目，通常數值越大越好。Memory 則是看存取讀寫的速度(MB/sec)。在刪除不必要功能並降低 kernel size 後，發現原先的 CPU 與 memory 表現較好，這部分的結果分析會在後面做一個討論。

B. Geekbench

- Before reducing kernel size

```
pi@raspberrypi:~/Geekbench-6.2.0-LinuxARMPreview $ ./geekbench6
Geekbench 6.2.0 Preview : https://www.geekbench.com/

Geekbench 6 for Linux/AArch64 is a preview build. Preview builds require an
active Internet connection and automatically upload benchmark results to the
Geekbench Browser.

System Information
  Operating System      Debian GNU/Linux 12 (bookworm)
  Kernel                Linux 6.1.93-v8+ aarch64
  Model                 Raspberry Pi 4 Model B Rev 1.4
  Motherboard           N/A

CPU Information
  Name                  ARM ARMv8
  Topology              1 Processor, 1 Core, 4 Threads
  Identifier            ARM implementer 65 architecture 8 variant 0 part
3336 revision 3
  Base Frequency        1.80 GHz

Memory Information
  Size                  7.63 GB
```

```
Uploading results to the Geekbench Browser. This could take a minute or two
depending on the speed of your internet connection.

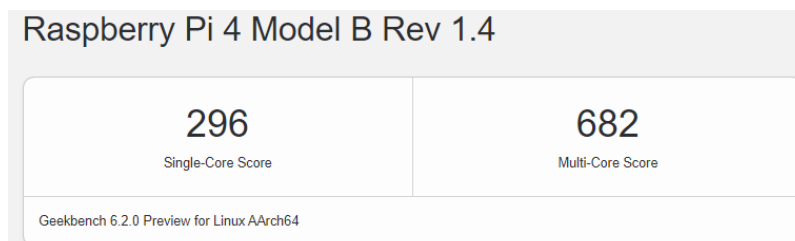
Upload succeeded. Visit the following link and view your results online:

  https://browser.geekbench.com/v6/cpu/7964713

Visit the following link and add this result to your profile:

  https://browser.geekbench.com/v6/cpu/7964713/claim?key=590426
```

測試結果會被放在一線上連結，如上圖所示，點進去後可以看到各項評分。
結果如下：



System Information	
System Information	
Operating System	Debian GNU/Linux 12 (bookworm)
Model	Raspberry Pi 4 Model B Rev 1.4
Motherboard	N/A
CPU Information	
Name	ARM ARMv8
Topology	1 Processor, 1 Core, 4 Threads
Identifier	ARM implementer 65 architecture 8 variant 0 part 3336 revision 3
Base Frequency	1.80 GHz
Cluster 1	0 Cores
Memory Information	
Size	7.63 GB

Single-Core Performance

Single-Core Score	296	
File Compression	309 44.4 MB/sec	
Navigation	411 2.48 routes/sec	
HTML5 Browser	371 7.59 pages/sec	
PDF Renderer	456 10.5 Mpixels/sec	
Photo Library	244 3.31 images/sec	
Clang	412 2.03 Klines/sec	
Text Processing	305 24.4 pages/sec	
Asset Compression	400 12.4 MB/sec	
Object Detection	93 2.77 images/sec	
Background Blur	186 0.77 images/sec	
Horizon Detection	444 13.8 Mpixels/sec	
Object Remover	184 14.2 Mpixels/sec	
HDR	371 10.9 Mpixels/sec	
Photo Filter	238 2.36 images/sec	
Ray Tracer	378 365.4 Kpixels/sec	
Structure from Motion	246 7.78 Kpixels/sec	

Multi-Core Performance

Multi-Core Score	682	
File Compression	321 46.1 MB/sec	
Navigation	759 4.57 routes/sec	
HTML5 Browser	799 16.4 pages/sec	
PDF Renderer	1333 30.7 Mpixels/sec	
Photo Library	742 10.1 images/sec	
Clang	1124 5.53 Klines/sec	
Text Processing	363 29.1 pages/sec	
Asset Compression	1513 46.9 MB/sec	
Object Detection	266 7.96 images/sec	
Background Blur	181 0.75 images/sec	
Horizon Detection	986 30.7 Mpixels/sec	
Object Remover	561 43.2 Mpixels/sec	
HDR	935 27.4 Mpixels/sec	
Photo Filter	589 5.64 images/sec	
Ray Tracer	1510 1.46 Mpixels/sec	
Structure from Motion	843 26.7 Kpixels/sec	

- After reducing kernel size

```
pi@raspberrypi:~/Geekbench-6.2.0-LinuxARMPreview $ ./geekbench6
Geekbench 6.2.0 Preview : https://www.geekbench.com/

Geekbench 6 for Linux/AArch64 is a preview build. Preview builds require an
active Internet connection and automatically upload benchmark results to the
Geekbench Browser.

System Information
  Operating System      Debian GNU/Linux 12 (bookworm)
  Kernel                Linux 6.1.93-v8+ aarch64
  Model                 Raspberry Pi 4 Model B Rev 1.4
  Motherboard           N/A

CPU Information
  Name                  ARM ARMv8
  Topology              1 Processor, 1 Core, 4 Threads
  Identifier             ARM implementer 65 architecture 8 variant 0 part
3336 revision 3
  Base Frequency        1.80 GHz

Memory Information
  Size                  7.63 GB

Single-Core
Running File Compression

Uploading results to the Geekbench Browser. This could take a minute or two
depending on the speed of your internet connection.

Upload succeeded. Visit the following link and view your results online:

  https://browser.geekbench.com/v6/cpu/7957737

Visit the following link and add this result to your profile:

  https://browser.geekbench.com/v6/cpu/7957737/claim?key=418293
```

結果一樣會被上傳在一個線上的連結，詳細結果如下:

Raspberry Pi 4 Model B Rev 1.4

291	643
Single-Core Score	Multi-Core Score

Geekbench 6.2.0 Preview for Linux AArch64

System Information

System Information	
Operating System	Debian GNU/Linux 12 (bookworm)
Model	Raspberry Pi 4 Model B Rev 1.4
Motherboard	N/A

CPU Information

Name	ARM ARMv8
Topology	1 Processor, 1 Core, 4 Threads
Identifier	ARM implementer 65 architecture 8 variant 0 part 3336 revision 3
Base Frequency	1.80 GHz
Cluster 1	0 Cores

Memory Information

Size	7.63 GB
------	---------

Single-Core Performance

Single-Core Score	291	
File Compression	299 43.0 MB/sec	
Navigation	408 2.46 routes/sec	
HTML5 Browser	349 7.15 pages/sec	
PDF Renderer	447 10.3 Mpxels/sec	
Photo Library	238 3.23 images/sec	
Clang	402 1.98 Klines/sec	
Text Processing	305 24.5 pages/sec	
Asset Compression	396 12.3 MB/sec	
Object Detection	91 2.72 images/sec	
Background Blur	179 0.74 images/sec	
Horizon Detection	436 13.6 Mpxels/sec	
Object Remover	180 13.8 Mpxels/sec	
HDR	370 10.8 Mpxels/sec	
Photo Filter	237 2.35 images/sec	
Ray Tracer	379 366.5 Kpxels/sec	
Structure from Motion	238 7.54 Kpxels/sec	

Multi-Core Performance

Multi-Core Score	643	
File Compression	307 44.1 MB/sec	
Navigation	729 4.39 routes/sec	
HTML5 Browser	765 15.7 pages/sec	
PDF Renderer	1276 29.4 Mpxels/sec	
Photo Library	707 9.59 images/sec	
Clang	982 4.84 Klines/sec	
Text Processing	358 28.6 pages/sec	
Asset Compression	1420 44.0 MB/sec	
Object Detection	246 7.37 images/sec	
Background Blur	166 0.69 images/sec	
Horizon Detection	970 30.2 Mpxels/sec	
Object Remover	520 40.0 Mpxels/sec	
HDR	856 25.1 Mpxels/sec	
Photo Filter	570 5.65 images/sec	
Ray Tracer	1409 1.36 Mpxels/sec	
Structure from Motion	761 24.1 Kpxels/sec	

以上為 `geekbench` 分析，`geekbench` 是一個開源的基礎測試工具，可以為系統的各項指標評分，也可以進行跨平台的處理與比較。在這次實驗中，也發現在刪除 `kernel` 不必要的功能後各項指標也都下降了。理論上若是刪除不必要的功能與模組後，效能理應提升因為 `CPU` 和 `memory` 的 `loading` 變小，然而我卻發現不論是在 `sysbench` 或是 `geekbench` 表現中都變差，我推測是因為在關閉的項目中有可能有些影響到硬體的表现，進而影響到效能。例如 `Power Management` 會直接關閉電源的各項功能管理，無法進行動態的功耗配置，反而使資源分配不佳，以上種種原因還有待後續研究。

3. Q3: Patch your kernel to support real-time tasks.

本部分則是 `patch` 的實作。`Patch` 又稱作補丁，通常是一段程式碼的修改，或是套件的更新，主要目的在不希望更改整個環境的情況下做一些細部更動。會以 `+`、`-` 的方式做修改，例如：`- int a = 0;`、`+ int a = 1;`，實驗需要尋找適當的 `kernel` 版本，更改 `kernel branch` 並找到在 <https://mirrors.edge.kernel.org/pub/linux/kernel/projects/rt/> 中能夠對應的 `patch`。

首先有尋找到 3.6 版本符合，然而後來發現無法 `build` 成功，因為 3.6 版本需要的編譯器版本為 `gcc4` 或 `gcc5`，而系統內建的版本為 `gcc11` 因此不相容而 `make` 失敗。

```
wang@EOS:~/Desktop/linux$ head -n 4 Makefile
VERSION = 3
PATCHLEVEL = 6
SUBLEVEL = 11
```

Index of /pub/linux/kernel/projects/rt/3.6

../	12-Sep-2013 00:28	-
incr/	14-Sep-2013 01:36	-
older/	13-Sep-2013 18:51	-
stable/	19-Nov-2013 22:01	2674
README	19-Nov-2013 22:01	490
README.sign	14-Sep-2013 01:36	134K
patch-3.6.11.9-rt42.patch.bz2	14-Sep-2013 01:36	168K
patch-3.6.11.9-rt42.patch.gz	14-Sep-2013 01:36	490
patch-3.6.11.9-rt42.patch.sign	14-Sep-2013 01:36	139K
patch-3.6.11.9-rt42.patch.xz	14-Sep-2013 01:36	235K
patches-3.6.11.9-rt42.tar.bz2	14-Sep-2013 01:36	323K
patches-3.6.11.9-rt42.tar.gz	14-Sep-2013 01:36	490
patches-3.6.11.9-rt42.tar.sign	14-Sep-2013 01:36	228K
patches-3.6.11.9-rt42.tar.xz	25-Apr-2023 21:13	1530
sha256sums.asc		

接著陸續試到 4.19.127 才成功

```
wang@EOS:~/Desktop/linux$ head Makefile -n 4
# SPDX-License-Identifier: GPL-2.0
VERSION = 4
PATCHLEVEL = 19
SUBLEVEL = 127
```

最後和前面步驟相同，產生 default config 檔重新 build kernel，再 flash image 到 SD 卡上即可。完成結果如下：

```
pi@raspberrypi:~ $ uname -r
4.19.127-rt54-v8+
```

另外，我也上網查了一些對於 real time tasks 的效能測試，利用 `cyclicttest` 可以檢測 kernel 對 threads 的延遲，進而判斷表現。測試如下：

```
pi@raspberrypi:~ $ sudo cyclicttest -t1 -p 80 -i 1000 -l 100000
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 1.71 0.97 0.40 1/206 6934
T: 0 ( 4197) P:80 I:1000 C: 100000 Min:      3 Act:      6 Avg:      6 Max:    1046
```

其中，T 代表 thread(PID)、-p 80 為優先等級、-i 1000 為時間間隔、-l 100000 為循環測試次數，結果表明最小延遲時間為 3 微秒、實際延遲時間與平均延遲時間為 6 微秒，但最大延遲時間為 1046 微秒需要比較注意，整體來說表現還算不錯。

4. Reference

- A. <https://linuxconfig.org/how-to-benchmark-your-linux-system>
- B. <https://www.linkedin.com/pulse/optimizing-linux-kernel-image-size-pranav-shivalingaiah-u5dec>
- C. <https://www.acontis.com/en/building-a-real-time-linux-kernel-in-ubuntu-preemptrt.html>
- D. https://blog.csdn.net/weixin_44723087/article/details/120048651
- E. https://www.raspberrypi.com/documentation/computers/linux_kernel.html#cross-compile-the-kernel